

Question 1 : Answer the following :

(4 M)

- a) The formal activities of *Operations Research* (OR) were initiated in ----- **England** during World War II.
- b) An optimization problem is called ----- (**LPP**) when the objective function and all the constraints are linear functions of the decision variables.
- c) To solve a LPP algebraically, we first put it in the **Standard Form**. What is this means?
This means:
1. All decision variables are **nonnegative**.
2. All constraints (with the exception of the non-negativity of the variables) are **equations with nonnegative RHS**.

Question 2 : Solve the following problem Graphically:

(6 M)

Maximum $z = 10x_1 + 5x_2$

S. to:

$x_1 = 4$

$x_2 = 3$

$z = 55$

$\rightarrow x_1 - x_2 \leq 1$

$3x_1 + x_2 \geq 7$

$\rightarrow x_2 \leq 3$

$x_1 \geq 0, x_2 \geq 0$

$x_1 - x_2 = 1$

$3x_1 + x_2 = 7$

$x_2 = 3$

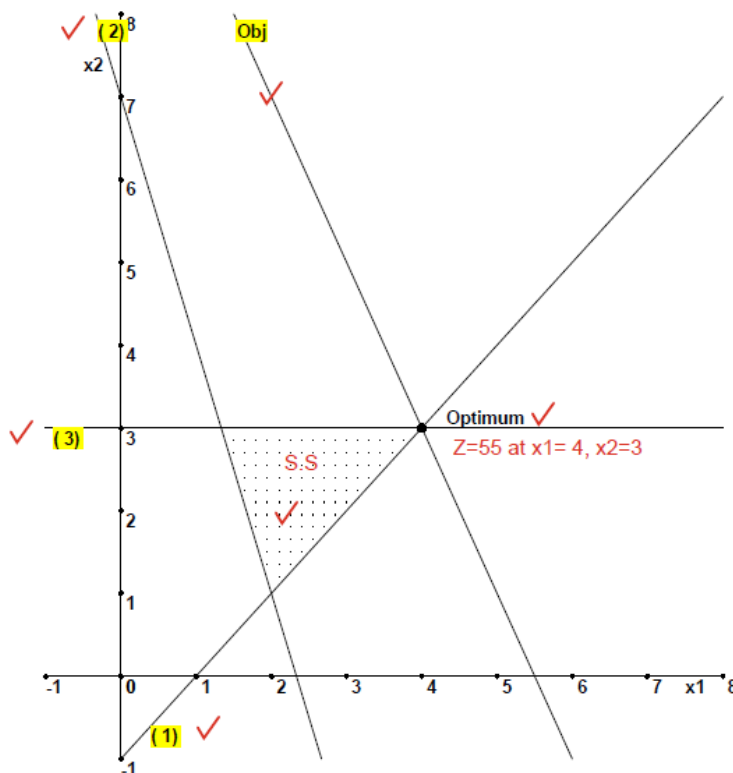
$\frac{1}{-1} \quad \frac{1}{1}$

$(0, -1) \quad (1, 0)$

$\frac{7}{1} \quad \frac{7}{3}$

$(0, 7) \quad (2.33, 0)$

$(0, 3)$



Question 3 : Given the following problem:**(6 M)**

Minimize $Z = 7x_1 + 3x_2$

S.to

$$2x_1 + x_2 \leq 8$$

$$x_1 - 3x_2 \geq 12$$

$$4x_2 = -3$$

$$x_1 \geq 0, x_2 \text{ Unrestricted in sign.}$$

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). **"Don't solve the problem"**

Solution:

1. Graphical method and Algebraic method. **1**
- 2.

Minimize $Z = 7x_1 + 3x_2^+ - 3x_2^- + 0s_1 + 0s_2$ **1**

S.to:

$$2x_1 + x_2^+ - x_2^- + s_1 = 8 \quad 1/2$$

$$x_1 - 3x_2^+ + 3x_2^- - s_2 = 14 \quad 1/2$$

$$-4x_2^+ + 4x_2^- = 3 \quad 1$$

$$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0 \quad 1/2$$

Question 4 : Given the following problem:**(9 M)**

Maximize $Z = 8x_1 + 7x_2 + 0s_1 + 0s_2 + 0s_3$

S.to

$$20x_1 + 16x_2 \leq 62 \quad +s_1 = 62$$

$$-x_1 + x_2 \leq 1 \quad +s_2 = 1$$

$$x_1 \leq 3 \quad +s_3 = 3$$

$$x_1 \geq 0, x_2 \geq 0 \quad x_1, x_2, s_1, s_2, s_3 \geq 0$$

- a) Write the problem in **standard form** (equation form).
- b) Solve the problem by the **simplex method** (stop at the first iteration **"Only two tables"**).

Solution:

a) Maximize $Z = 8x_1 + 7x_2 + 0s_1 + 0s_2 + 0s_3$ **1/2**

S.to:

$$20x_1 + 16x_2 + s_1 = 62 \quad 1/2$$

$$-x_1 + x_2 + s_2 = 1 \quad 1/2$$

$$x_1 + s_3 = 3 \quad 1/2$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0 \quad 1/2$$

b)

	x1	x2	S1	S2	S3	Sol.
Z	-8	-7	0	0	0	0
S1	20	16	1	0	0	62
S2	-1	1	0	1	0	1
S3	1	0	0	0	1	3
Z	0	-7	0	0	8	24
S1	0	16	1	0	-20	2
S2	0	1	0	1	1	4
x1	1	0	0	0	1	3

First table 2m (every row 0.5m + 0.5m for pivot element)

Second table 4m (1m for every row) , Every mistake in calculation – 0.5m)